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```
import heapq

graph = {
    'A': [('B', 4), ('C', 2)],
    'B': [('A', 4), ('C', 1), ('D', 5)],
    'C': [('A', 2), ('B', 1), ('D', 8)],
    'D': [('B', 5), ('C', 8)]
}

def dijkstra(start, goal):
    queue = [(0, start, [start])]
    visited = set()

    while queue:
        cost, node, path = heapq.heappop(queue)

        if node in visited:
            continue

        visited.add(node)

        if node == goal:
            print("Shortest Path:", path)
            print("Minimum Transportation Cost:", cost)
            return

        for neighbor, weight in graph[node]:
            if neighbor not in visited:
                heapq.heappush(queue, (cost + weight, neighbor, path + [neigh

    print("No path found")

dijkstra('A', 'D')
```

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```
Python 3.13.1 (tags/v3.13.1:0671451, Dec  3 2024, 19:06:28) [MSC v.1942
64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
========== RESTART: C:/Users/user/Downloads/AI.py ============
========
Shortest Path: ['A', 'C', 'B', 'D']
Minimum Transportation Cost: 8
>>>
```